Madhavan Srajan Gupta

Android Software Engineer

Pune, Maharashtra, India | Open to Bengaluru (Hybrid/Remote) | linkedin.com/in/msg17 | github.com/Madhavan1707

SUMMARY

Android Software Engineer with 2.5+ years of experience building safety-critical and high-performance systems for the automotive telematics industry. Core expertise in Android Java/Kotlin framework development, real-time vehicle connectivity (MQTT, Protobuf, AES encryption), and inter-process communication (AIDL, background services). Demonstrated track record delivering sub-50ms latency systems, 20% crash reduction, and safety-critical compliance features (eCall, bCall, SOS) deployed on Quectel telematics modules at Visteon Corporation. Additional background in distributed data engineering (PySpark, Kafka, AWS EMR, Databricks). Seeking a Senior Android Developer or Automotive Software Engineer role where reliability and innovation matter.

SKILLS

Android & Mobile: Java, Kotlin, Android SDK, Android Studio, AIDL, Background Services, Broadcast Receivers, MVVM, Room, Retrofit, RxJava2, Dagger, Firebase, Glide, Jetpack, JUnit, Espresso

Automotive & Telematics: MQTT, Protobuf, AES-256 Encryption, ZSTD Compression, CAN Bus, Quectel Modules, TCU Integration, eCall/bCall/SOS, Real-Time Systems, Safety-Critical Systems, IPC

Data Engineering: PySpark, Apache Spark, Apache Kafka, AWS EMR, Databricks, ETL Pipelines, SQL, Pandas

Tools & Platforms: Git, Android Automotive OS, Linux, REST APIs, JSON, Agile/Scrum, Jira

EXPERIENCE

Software Engineer 2 — Android Telematics Sep 2024 – Present

Visteon Corporation · Pune, India

- Architected and delivered safety-critical Emergency Call (eCall), Breakdown Call (bCall), SOS, and Panic Alert features in Android Java on Quectel chipsets, ensuring EU automotive regulatory compliance for next-generation vehicle platforms
- Designed the Unified Communication Layer (UCL) integrating Android services, AIDL interfaces, and broadcast receivers, reducing inter-process communication latency by ~30% in safety-critical execution paths
- Built high-availability background Android services for continuous TCU-to-backend communication, maintaining 99.9%+ uptime in production validation environments
- Authored end-to-end technical design documentation (sequence diagrams, state machine flows, API specs) adopted as the team standard across 3+ concurrent feature tracks
- Partnered with firmware and hardware teams on Quectel module integration, compressing feature validation cycles from 3 weeks to under 10 days

Software Engineer — Vehicle Connectivity Feb 2024 – Sep 2024

Visteon Corporation · Pune, India

- Led development of a real-time vehicle tracking system in Android Java achieving sub-50ms end-to-end latency via MQTT, Protobuf serialization, AES-256 encryption, and ZSTD compression
- Reduced data bandwidth consumption by ~35% through ZSTD compression integration without sacrificing transmission reliability
- Developed Android background services for CAN bus signal monitoring that maintained continuous data capture during low-power vehicle states, eliminating data loss events in all field testing scenarios
- Validated end-to-end communication flows with backend and network teams, producing a deployment-ready architecture for fleet-scale telematics

Software Engineer — Android Telematics POC Aug 2023 – Feb 2024

Visteon Corporation · Pune, India

- Engineered the Android telematics layer for a two-wheeler POC, including MQTT pub/sub, AIDL interfaces, JSON payload handlers, event-driven location tracking, and heartbeat monitoring modules
- Built multi-threaded background services managing timers, handlers, and dynamic service binding, achieving stable performance under severe memory constraints in two-wheeler environments
- Delivered POC demonstration 2 weeks ahead of schedule by independently owning end-to-end Android framework integration

Data Engineer Intern Jan 2023 – Aug 2023

AB InBev GCC India · Bengaluru, India

- Designed and deployed distributed ETL pipelines in PySpark processing 5+ TB of daily data; reduced pipeline execution time by 40% via Spark cluster tuning on AWS EMR and Databricks
- Built schema validation and data cleansing routines (PySpark DataFrame API + Pandas) reducing downstream data quality incidents by 60%
- Integrated streaming data from Apache Kafka, REST APIs, and relational databases into a centralized data lake, enabling near-real-time analytics for production business dashboards

PROJECTS

Real-Time Vehicle Telematics Connectivity Framework

Production-grade Android Java telematics framework with MQTT message broker integration, Protobuf-based serialization, AES-256 payload encryption, and ZSTD compression. Achieved sub-50ms vehicle-to-cloud latency. Architecture supports fleet-scale deployments.

Stack: Android Java, MQTT, Protobuf, AES-256, ZSTD, Background Services, AIDL

Safety-Critical Automotive Communication System

Developed the emergency call stack (eCall, bCall, SOS, Panic Alert) for an automotive telematics platform on Quectel modules, integrating Android framework services with TCU hardware and backend emergency services infrastructure.

Stack: Android Java, AIDL, Broadcast Receivers, UCL, Quectel SDK

Trending Movies Application | Personal Project | Apr 2025 – Jun 2025

Android app consuming TMDB REST API, demonstrating clean architecture with MVVM, Dagger DI, Room local persistence, RxJava2 reactive streams, and Glide image loading.

Stack: Android Java, MVVM, Dagger, Room, RxJava2, Retrofit, Glide, Android Studio

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science 2019 – 2023

Vellore Institute of Technology (VIT)